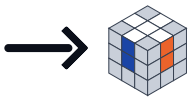
# 1 Cross





F



F2

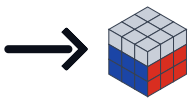


R' F R



F D' L D

# 2 F2L #1 – #24



### Four Basic Cases



#1
F' U' F



#2
R U R'



#3
(U R U' R')



#4
U' F' U F

### Both pieces on U



#5
F' U F U2
R U R'



#6
U' R U2 R'
U2 R U' R'



#7
F U2 F2 U'
F2 U' F'



#8
F' U2 F
U F' U' F



#9
U2 R U
R' (U R U' R')



#10
U' F' U2
F2 R' F' R



#11
U R U2
R' (U R U' R')



#12
U2 R U R'
F' U' F



#13
R U2 R'
U' R U R'



#14
R' U2 R2
U R2 U R



#15
R2 U2 F R2
F' U2 R2



#16
R U' R'
U2 F' U' F



#17
U F' U F
U' F' U' F



#18
U' R U R'
U2 R U' R'



#19
U R U R'
U2 F' U' F



#20
U2 R2 U2 R'
U' R U' R2



#21
U2 F2 U2 F
U F' U F2



#22
U' R U R'
U R U R'

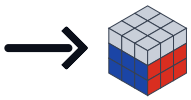


#23
U F' U' F
U F' U2 F

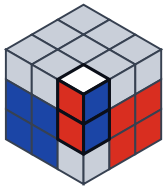


#24
U' R U' R'
U R U R'

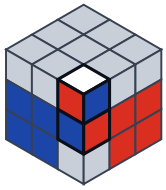
# 2 F2L #25 – #41



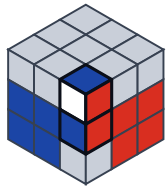
### Edge in slot



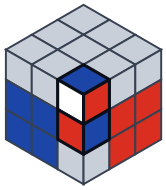
#25
R U' R'
F' U2 F



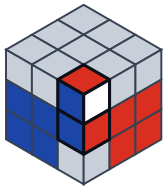
#26
R2 U R2 U
R2 U2 R2



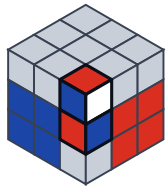
#27
U' R U' R'
U2 R U' R'



#28
U' R U R'
U F' U' F

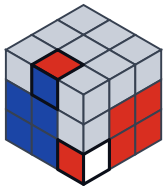


#29
U R U R'
U2 R U R'

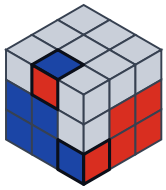


#30
U F' U' F
U' R U R'

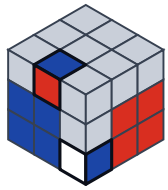
### Corner in slot



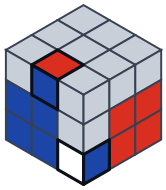
#31
R U2 R'
F' U2 F



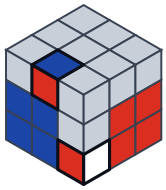
#32
F' U2 F
U2 R U R'



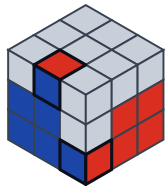
#33
U' R U'
R2 F R F'



#34
U R' F R
F2 U' F

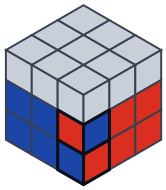


#35
U F' U F
R U R'

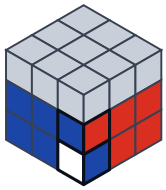


#36
U
(R U R' U')
F' U' F

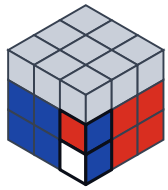
### Both in slot (incorrect)



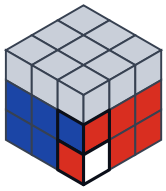
#37
R U' R
U2 F R2
F' U2 R2



#38
R U' R' U'
R U R' U2
R U' R'



#39
R F
(U R U' R')
F' U' R'

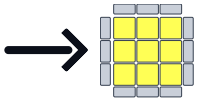


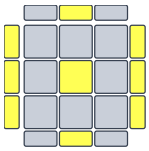
#40
R U2 R U R'
U R U2 R2



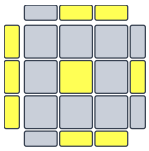
#41
R U F
(R U R' U')
F' R'

# 3 OLL #1 – #19

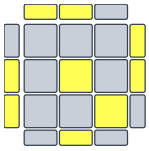




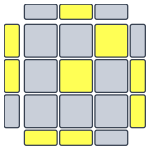
#1 Dots
R U2 R2 F R F'
U2 (R' F R F')



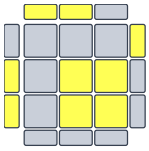
#2 Dots
F (R U R' U') F'
f (R U R' U') f'



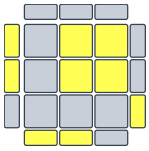
#3 Dots
f (R U R' U') f'
U' F (R U R' U') F'



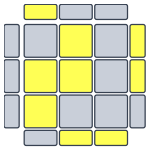
#4 Dots
f (R U R' U') f'
U F (R U R' U') F'



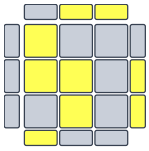
#5 Squares
r' U2 R U R' U r



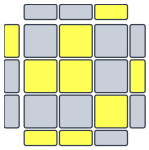
#6 Squares
r U2 R' U' R U' r'



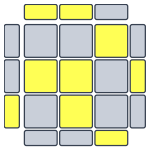
#7 Lightning
r U R' U R U2 r'



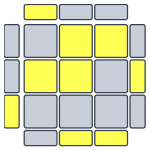
#8 Lightning
r' U' R U' R' U2 r



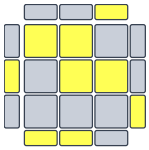
#9 Fish
(R U R' U') R'
F R2 U R' U' F'



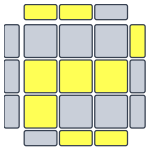
#10 Fish
R U R' U
(R' F R F') R U2 R'



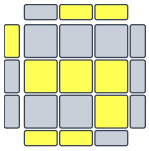
#11 Lightning
r U R' U
(R' F R F') R U2 r'



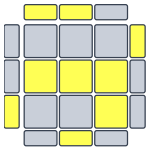
#12 Lightning
M' R' U' R U'
R' U2 R U' M



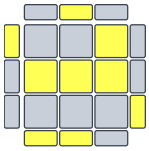
#13 Knight
F U R U' R2 F'
R (U R U' R')



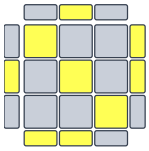
#14 Knight
R' F R U R' F' R F U' F'



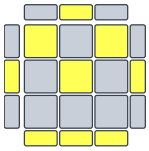
#15 Knight
r' U' r R' U' R U r' U r



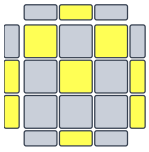
#16 Knight
r U r'
(R U R' U') r U' r'



#17 Dots
F R' F' R2 r'
(U R U' R') U' M'

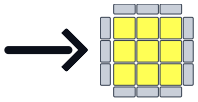


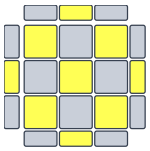
#18 Dots
r U R' U R U2 r2
U' R U' R' U2 r



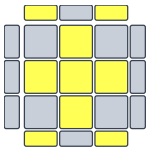
#19 Dots
r' R U (R U R' U') M'
(R' F R F')

# 3 OLL #20 – #38

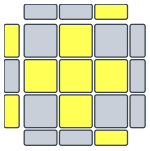




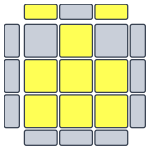
#20 Dots
r U R' U' M2
(U R U' R') U' M'



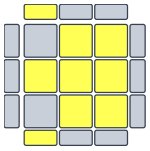
#21 Cross
R U2 R' U'
(R U R' U') R U' R'



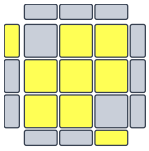
#22 Cross
R U2 R2 U' R2 U' R2 U2 R



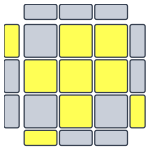
#23 Cross
R2 D' R U2 R' D R U2 R



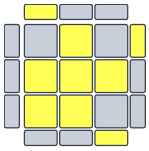
#24 Cross
r U R' U' r' F R F'



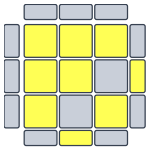
#25 Cross
F' r U R' U' r' F R



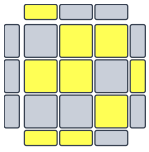
#26 Cross
(R U2 R' U' R U' R')



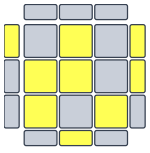
#27 Cross
(R U R' U R U2 R')



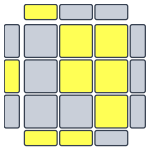
#28 Edges only
r U R' U' r'
R (U R U' R')



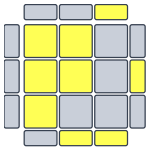
#29 Awkward
(R U R' U') R U'
R' F' U' F R U R'



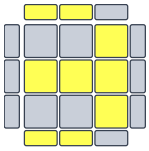
#30 Awkward
F U R U2 R' U'
R U2 R' U' F'



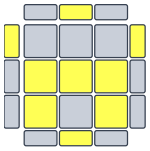
#31 P
R' U' F (U R U' R') F' R



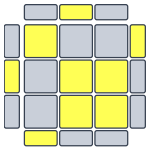
#32 P
L U F' U' L' U L F L'



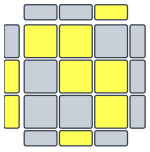
#33 T
(R U R' U') (R' F R F')



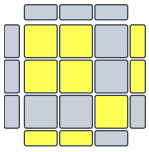
#34 C
R U R2 U' R'
F R U R U' F'



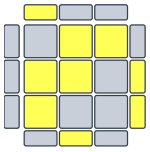
#35 Fish
R U2 R2 F R F' R U2 R'



#36 W
L' U' L U' L'
U L U L F' L' F

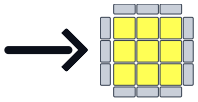


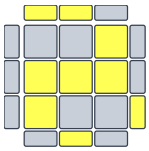
#37 Fish
(F R' F' R) (U R U' R')



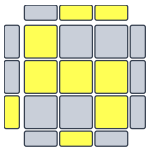
#38 W
R U R' (U R U' R') U'
(R' F R F')

# 3 OLL #39 – #57

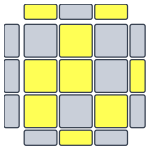




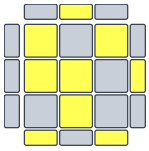
#39 Lightning
L F' L' U' L U F U' L'



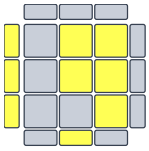
#40 Lightning
R' F (R U R' U') F' U R



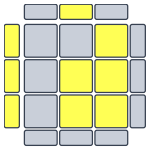
#41 Awkward
(R U R' U R U2 R')
F (R U R' U') F'



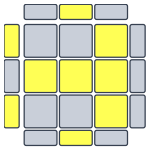
#42 Awkward
R' U' R U' R' U2 R
F (R U R' U') F'



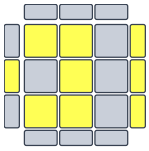
#43 P
F' U' L' U L F



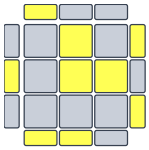
#44 P
f (R U R' U') f'



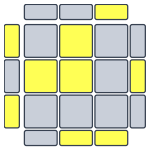
#45 T
F (R U R' U') F'



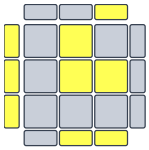
#46 C
R' U' (R' F R F') U R



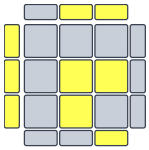
#47 Large L
F' L' U' L U L' U' L U F



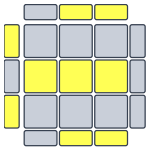
#48 Large L
F (R U R' U')
(R U R' U') F'



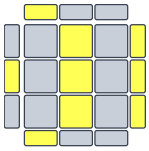
#49 Large L
r U' r2 U r2 U r2 U' r



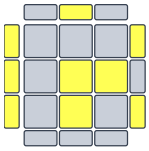
#50 Large L
r' U r2 U' r2 U' r2 U r'



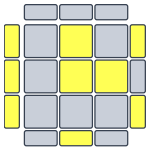
#51 Lines
f (R U R' U')
(R U R' U') f'



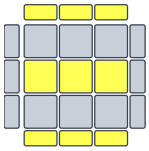
#52 Lines
R' U' R U' R' U F' U F R



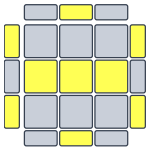
#53 Large L
r' U' R U' R'
(U R U' R') U2 r



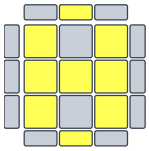
#54 Large L
r U R' (U R U' R') U
R U2 r'



#55 Lines
R' F R U R U' R2 F'
R2 U' R' U R U R'

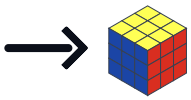


#56 Lines
r' U' r U' R'
(U R U' R') U R r' U r



#57 Edges only
(R U R' U') M' U R U' r'

# 4 PLL

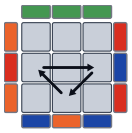


**Headlights?**





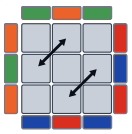

**Ua**
R U' R U R
(U R U' R') U' R2



**Ub**
R2 U (R U R' U') R'
U' R' U R'

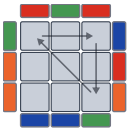


**H**
M2 U M2 U2 M2 U M2

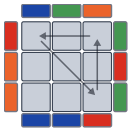


**Z**
M2 U M2 U M'
U2 M2 U2 M' U2

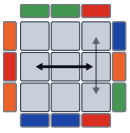


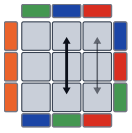
**Aa**
x R' U R' D2 R
U' R' D2 R2 x'



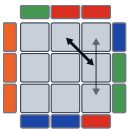
**Ab**
x R2 D2 R U R'
D2 R U' R x'



**T**
(R U R' U') R' F R2
U' R' U' R U R' F'



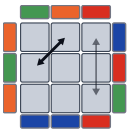
**F**
R' U' F' (R U R' U') R'
F R2 U' R' U' R U R' U R



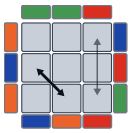
**Ja**
x R2 F R F' R
U2 r' U r U2 x'



**Jb**
R U R' F' (R U R' U') R'
F R2 U' R' U'



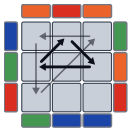
**Ra**
R U' R' U' R U R D R'
U' R D' R' U2 R' U'



**Rb**
R2 F R (U R U' R') F'
R U2 R' U2 R U



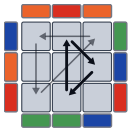
**Ga**
R2 U R' U R' U' R U'
R2 U' D R' U R D' U



**Gb**
U' R' U' R U D' R2 U
R' U R U' R U' R2 D



**Gc**
R2 U' R U' R U R' U
R2 U D' R U' R' D U

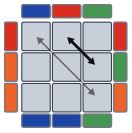


**Gd**
U' (R U R' U') D R2 U'
R U' R' U R' U R2 D'





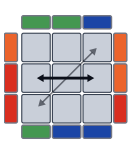
**E**
x' R U' R' D R U R' D'
R U R' D R U' R' D' x



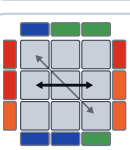
**V**
R U2 R' D R U' R U'
R U R2 D R' U' R D2



**Y**
F R U' R' U' R U R' F'
(R U R' U') (R' F R F')



**Na**
R U R' U R U R' F'
(R U R' U') R' F
R2 U' R' U2 R U' R'



**Nb**
R' (U R U' R') F' U' F
R U R' (F R' F' R) U' R